

Arnab Chakraborty

Full Stack Developer

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Professional Summary

Full Stack Developer with hands-on experience building full stack, front-end web, and back-end systems using Next.js, React, TypeScript, Node.js, and Python. Strong in web design, problem solving, and organizational skills, with measurable impact across open-source and production projects.

Education

Gauhati University, Bachelor of Technology (B.Tech.) in Information Technology Aug 2023 - May 2027
(Expected)

Relevant Coursework: Computer Science fundamentals, Data Structures, OOP, DBMS, Operating Systems

Experience

Open-Source Contributor | MetaBrainz Picard Mar 2024 - Present

- Implemented a feature to show old and new file paths during renaming, reducing user confusion by 30%. (PR #2563)
- Built a custom title-casing function fixing 100+ capitalization issues and improving metadata consistency. (PR #2548)
- Added Python unit tests with Pytest, increasing test coverage by 10%.
- **Technologies:** Git, GitHub, Python, PyQT, Pytest

Projects

InvenTrack - 2025 inventrack.zytact.com

- Developed a full-stack inventory system with AI/LLM-powered forecasting and recommendations using Gemini API.
- Built Python preprocessing pipelines for inventory trend analysis and context-aware prompts.
- Designed a multi-tenant database with Drizzle ORM and automated reorder suggestions.
- **Technologies:** Python, Next.js, Clerk, TypeScript, Tailwind CSS, Shadcn UI, TanStack Query, Drizzle ORM, Gemini API

Compress - 2026 compress.zytact.com

- Built a privacy-first in-browser image compressor using WebAssembly.
- Implemented Lanczos resampling and target-size compression.
- Delivered 100% client-side processing with no server uploads.
- **Technologies:** Rust, WebAssembly, TypeScript

Strokeshare - 2025 strokeshare.zytact.com

- Built an infinite-canvas drawing app with export, undo/redo, and persistence.
- Implemented performant state management with Zustand.
- **Technologies:** Next.js, TypeScript, Tailwind CSS, React Konva, Zustand

Technical Skills

Programming Languages: Python, C, JavaScript, TypeScript, HTML, CSS

AI / Machine Learning: Generative AI, Large Language Models (LLMs), Prompt Engineering, Hugging Face Transformers, OpenAI API, Gemini API, NumPy, Pandas

Front End Web: React.js, Next.js, HTML, CSS, JavaScript, TypeScript, Tailwind CSS, TanStack Query, Zustand, React Konva, Web design

Back End: Node.js, Express.js, Python, RESTful APIs, tRPC, Auth.js, Better Auth, Clerk

Full Stack: End-to-end web application development, API integration, authentication, deployment-ready architecture

Database: MongoDB, PostgreSQL, MySQL

Exposure: AngularJS, PHP, .NET, Ruby, Oracle

Tools: Git, GitHub Actions, Linux, Vim/NeoVim, Jest, Vitest, Pytest, ESLint, Prettier